

Smart Traffic-Light System Using Artificial Intelligence and Computer Vision for Emergency Vehicle Management

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Abstract—Traffic congestion and delayed emergency vehicle movement are important challenges in urban transportation systems. Conventional traffic lights generally operate using fixed or predefined timing schedules and may not respond efficiently to rapidly changing traffic conditions. This paper proposes a smart traffic-light system that uses artificial intelligence (AI), computer vision, and local edge-based processing to monitor traffic conditions and identify emergency vehicles. Camera data can be processed locally to estimate vehicle density, queue length, and emergency vehicle presence. Based on the detected conditions, the system can dynamically modify traffic-signal phases to provide a clearer route for emergency vehicles while maintaining normal traffic operation. The proposed approach aims to reduce unnecessary waiting time, improve traffic-signal adaptability, and support faster emergency vehicle movement.

Index Terms—Artificial Intelligence, Computer Vision, Smart Traffic Lights, Emergency Vehicles, Traffic Signal Control, Edge Computing, Machine Learning

I. INTRODUCTION

Urban traffic congestion is a major challenge in modern transportation systems. Increasing numbers of vehicles, unpredictable traffic conditions, and fixed traffic-signal timings can result in unnecessary waiting time at road intersections. The problem becomes more important when emergency vehicles such as ambulances and fire engines need to pass through congested roads.

Traditional traffic signals commonly use predefined timing schedules. Although these schedules are simple to implement, they may not respond effectively to changes in real-time traffic density. Artificial intelligence and machine-learning techniques provide an opportunity to develop more adaptive traffic-signal systems.

Deep Reinforcement Learning (DRL) has been investigated for intelligent traffic-signal control. Rasheed et al. reviewed the application of DRL to traffic signal control and discussed its ability to address complex and changing traffic conditions [1].

Computer vision provides another approach for obtaining real-time traffic information. Camera-based systems can identify vehicles and estimate traffic characteristics such as vehicle count and queue length. Gorodokin et al. proposed a machine-vision-based approach for adaptive traffic-light control, where

traffic information obtained from cameras can be used to modify signal phases and cycle durations [2].

The proposed research combines these concepts with local processing for emergency vehicle management. A smart camera system can continuously observe the intersection, identify traffic conditions, detect emergency vehicles, and provide suitable control information to the traffic-signal controller.

II. PROBLEM STATEMENT

Conventional traffic signals generally depend on fixed or predetermined timings. Such systems may not efficiently handle sudden variations in traffic density or provide a coordinated response when an emergency vehicle approaches an intersection.

Therefore, there is a need for an intelligent traffic-signal system capable of:

- Monitoring traffic conditions in real time.
- Estimating vehicle density and queue length.
- Detecting emergency vehicles.
- Dynamically modifying traffic-signal phases.
- Processing camera information locally with low communication delay.
- Providing a clearer route for emergency vehicles.

III. OBJECTIVES

The major objectives of the proposed system are:

- 1) To develop a smart camera-based traffic monitoring system.
- 2) To estimate traffic density using computer-vision techniques.
- 3) To identify emergency vehicles from camera input.
- 4) To dynamically control traffic-signal timing based on traffic conditions.
- 5) To provide priority to emergency vehicles at intersections.
- 6) To investigate the feasibility of local edge-based processing.

IV. LITERATURE REVIEW

Smart traffic-light systems are being studied as a way to improve traffic management by responding to changing road conditions. Traditional traffic signals generally operate using predefined timings, which may not respond effectively when traffic density changes significantly. Artificial intelligence (AI) and machine-learning techniques provide an opportunity to make traffic-signal control more adaptive by using information about current traffic conditions.

The reviewed studies demonstrate the potential of artificial intelligence, deep reinforcement learning, and machine vision for adaptive traffic-signal control. However, existing systems work well for regular daily traffic but lack a built-in way to automatically clear a path for a speeding ambulance during an emergency. This creates an opportunity to investigate a smart-camera-based approach that can detect emergency vehicles and provide priority while maintaining adaptive traffic-signal operation.

Therefore, the proposed work focuses on a local smart-camera-based system that can identify traffic conditions and emergency vehicles and assist in coordinating traffic signals to provide a clearer route for emergency vehicles.

V. PROPOSED SYSTEM

The proposed system consists of a smart camera, local processing unit, traffic analysis module, emergency vehicle detection module, and traffic signal controller.

The camera continuously captures traffic information at the intersection. The local processing unit analyzes the captured frames and estimates vehicle-related parameters.

The system can be divided into the following major modules:

- **Camera Module:** Captures real-time traffic images or video.
- **Vehicle Detection Module:** Identifies vehicles present in the camera view.
- **Traffic Density Module:** Estimates the number of vehicles and queue length.
- **Emergency Vehicle Detection Module:** Identifies emergency vehicles using visual features or a trained detection model.
- **Decision Module:** Determines the appropriate signal phase.
- **Traffic Signal Controller:** Applies the selected signal timing.

VI. SYSTEM FLOWCHART

VII. SYSTEM ARCHITECTURE

The proposed architecture is based on local processing near the traffic intersection. The camera provides real-time visual information to the local processing unit. The processing unit performs vehicle detection and traffic analysis before sending control information to the traffic-signal controller.

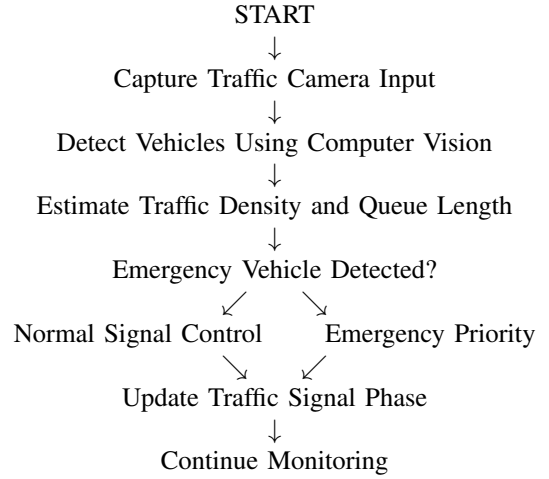


Fig. 1. Flowchart of the proposed smart traffic-light system.

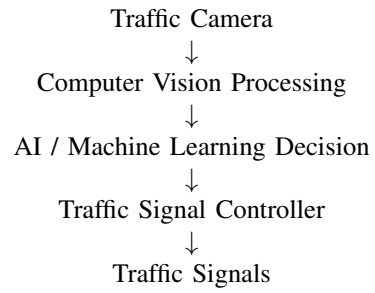


Fig. 2. Architecture of the proposed system.

VIII. MATHEMATICAL MODEL

The traffic density can be estimated using the number of detected vehicles within a particular observation region.

$$D = \frac{N_v}{A} \quad (1)$$

where D represents the estimated traffic density, N_v is the number of detected vehicles, and A represents the observation area.

The average queue length can be represented as:

$$Q = \frac{1}{N_v} \sum_{i=1}^{N_v} d_i \quad (2)$$

where Q represents the estimated average queue length and d_i represents the distance of the i -th vehicle from the intersection.

For emergency vehicle detection, a binary decision variable can be defined as:

$$E = \begin{cases} 1, & \text{if an emergency vehicle is detected,} \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

A simplified priority decision can then be represented as:

$$P = \begin{cases} P_{\text{high}}, & E = 1, \\ P_{\text{normal}}, & E = 0. \end{cases} \quad (4)$$

TABLE I
LITERATURE REVIEW AND RESEARCH GAP

Reference	Method / Contribution	Findings	Research Gap
Rasheed et al. [1]	Presented a comprehensive review of Deep Reinforcement Learning (DRL) for traffic signal control. The study discusses different DRL architectures, simulation platforms, complexity considerations, and design issues associated with intelligent traffic-signal control.	DRL can provide a more flexible alternative to fixed-time traffic-signal strategies and can handle complex and changing traffic conditions.	Existing DRL-based traffic-control approaches focus mainly on improving traffic-signal operation and do not provide a built-in mechanism to automatically clear a path for a speeding ambulance during an emergency.
Gorodokin et al. [2]	Proposed a machine-vision-based approach for adaptive traffic-light control. Camera-based traffic information is used to estimate traffic-flow characteristics and vehicle-related parameters near intersections.	Real-time visual information can support adaptive traffic-signal operation by dynamically adjusting signal phases and cycle durations.	The approach focuses on adaptive traffic control based on traffic information, but does not directly address automatic emergency-vehicle priority and path clearance for an ambulance.
Chu et al. [3]	Investigated traffic signal control using end-to-end off-policy deep reinforcement learning for intelligent traffic management.	Deep reinforcement learning can be used to learn traffic-signal control policies and improve adaptive signal operation.	Existing intelligent signal-control approaches do not directly combine adaptive traffic management with automatic ambulance path clearance during an emergency.

The signal-control decision may therefore consider traffic density, queue length, and emergency vehicle presence:

$$S = f(D, Q, E) \quad (5)$$

where S represents the selected traffic-signal state.

IX. ADAPTIVE SIGNAL CONTROL

The proposed system operates differently depending on the detected traffic condition.

TABLE II
ILLUSTRATIVE SIGNAL-CONTROL CONDITIONS

Condition	State	Action
Low density	Low	Normal timing
Medium density	Moderate	Adaptive timing
High density	High	Extended green
Emergency	Detected	Priority control

The table represents the conceptual operation of the system. Actual signal timings would depend on the intersection layout, traffic rules, safety constraints, and implementation environment.

X. EMERGENCY VEHICLE PRIORITY

When an emergency vehicle is detected, the system can identify its approach direction and communicate the required priority state to the traffic-signal controller.

A simplified sequence can be represented as:

- 1) Detect the approaching emergency vehicle.
- 2) Determine the direction of approach.
- 3) Verify the detection using multiple frames.
- 4) Determine the current traffic-signal state.
- 5) Select an appropriate priority phase.
- 6) Allow traffic movement in the emergency vehicle's direction.
- 7) Restore normal signal operation after the vehicle passes.

The system should include safety constraints so that conflicting traffic movements are not simultaneously permitted.

XI. ADVANTAGES

The proposed approach provides the following potential advantages:

- Real-time traffic monitoring.
- Adaptive signal operation.
- Detection of emergency vehicles.
- Local processing of camera information.
- Potential reduction in unnecessary data transmission.
- Support for faster decision-making at intersections.
- Ability to integrate computer vision with traffic-signal control.

XII. LIMITATIONS

Several practical limitations need to be considered:

- Camera performance may be affected by poor lighting and weather.
- Vehicle detection accuracy depends on the trained model and camera quality.
- Occlusion can make vehicle identification difficult.
- Emergency vehicle detection may produce false positives or false negatives.
- Real-world deployment requires compatibility with existing traffic controllers.
- Safety validation is required before deployment at public intersections.

XIII. EXPECTED OUTCOME

The proposed system is expected to demonstrate that real-time camera information can be used to support adaptive traffic-signal control. Emergency vehicle detection can be incorporated into the decision-making process so that the traffic signal can provide an appropriate priority phase.

The proposed local-processing approach may also reduce dependence on continuous transmission of video data to remote servers. Performance can be evaluated using parameters such as vehicle detection accuracy, emergency vehicle detection accuracy, average waiting time, queue length, and signal response time.

XIV. FUTURE SCOPE

Future work can investigate the following areas:

- Training advanced object-detection models for emergency vehicles.
- Integration with multiple intersections.
- Development of vehicle-to-infrastructure communication.
- Reinforcement-learning-based signal optimization.
- Integration with GPS-based emergency vehicle information.
- Testing using real-world traffic datasets.
- Deployment using edge-computing hardware.
- Development of a complete traffic-signal simulation environment.

XV. CONCLUSION

This paper presents a conceptual smart traffic-light system based on artificial intelligence, computer vision, and local processing. Unlike traditional fixed-time traffic signals, the proposed approach uses real-time traffic information to support adaptive signal operation.

The literature indicates that both deep reinforcement learning and machine-vision techniques can contribute to intelligent traffic-signal control [1]–[3]. Building on these approaches, the proposed system introduces emergency vehicle detection as an additional decision factor.

The system can monitor traffic, estimate vehicle density, detect emergency vehicles, and provide appropriate signal-control information. Further experimental development and real-world validation are required to determine the effectiveness, reliability, and safety of the proposed approach.

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